# Mahmoud Hafez

Data and Risk Analyst

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## / PROFILE

Data analyst with three years in risk and trading operations at a proprietary trading firm. I analyse trader behaviour, automate reviews and reporting, and build detection tools in Python and SQL. Equally comfortable answering one-off data questions from across the business, from a quick SQL pull to a full breakdown. Lead a back-office and risk team.

## / TOOLS

| | |
|---|---|
| **Python** | Pandas, NumPy, Matplotlib, Plotly, Streamlit, web scraping with BeautifulSoup |
| **SQL** | PostgreSQL, ClickHouse |
| **BI and reporting** | Tableau, Power BI, Excel, R and R Markdown |
| **Trading platforms** | MetaTrader 5, MatchTrader |
| **Risk analysis** | Trading-rule monitoring, copy-trading and hedging detection, linked-account checks |
| **Automation** | Automated reviews, scheduled reports, formatted Excel exports, internal dashboards |

## / EXPERIENCE

### Risk Analyst, Trading Operations

Nov 2023 to present

City Traders Imperium, remote

- Monitor trader behaviour across funded and evaluation accounts against the firm's trading rules.
- Lead the back-office and risk team, track SLAs and productivity, and report to management every week.
- Build the Python and SQL tooling that automates account reviews, detection and reporting (see below).
- Answer one-off data questions from across the business, from quick SQL pulls to full breakdowns.
- Process account upgrades and withdrawals on MT5 and MatchTrader, and escalate technical issues to IT.

### First Lieutenant

Sep 2020 to Apr 2023

Egyptian Air Defense Forces

## / WORK AT CITY TRADERS IMPERIUM

**Automated trading-rule review** — **90% faster**

Designed and built an automated review covering risk per trade, risk per trade idea, hedging, copy trading and the firm's other rules. Cut review time from about 20 minutes to 2 while checking more rules.

**Group trading detection** — **3 minutes**

Designed and built a detector that compares any number of traders, scores how closely each pair is connected and delivers an evidence report in about 3 minutes.

**Linked-account detection** — **3 minutes**

Designed linked-account detection and co-developed suspicious-connection checks, both now part of the automated account review. A full review averages about 3 minutes.

**Weekly management reporting** — **about 15 min**

Automated the weekly management report with a Python and SQL pipeline that produces a formatted Excel report in about 15 minutes.

## / PERSONAL PROJECTS

- **Sales analysis, answered in six tools.** A year of sales across 12 CSV files, analysed in Excel, SQL, R, Python, Tableau and Power BI, with recommendations on ad timing, promotion and stock.
- **What drives infant mortality in high-GDP countries.** A Tableau data story comparing infant mortality with health spending and birth rate, 2000 to 2012.
- **Top-rated businesses in the Yelp dataset.** SQL analysis of the top-rated restaurants, shops and spas across several states.

## / EDUCATION

### BSc Computer Science, Software Engineering

2015 to 2019

Mansoura University, Egypt

## / LANGUAGES

Arabic (native), English (fluent)

## / CERTIFICATES

- Google Data Analytics, Google
- Python for Everybody, University of Michigan
- SQL for Data Science, UC Davis
- Data Visualization with Tableau, UC Davis
- Python for Data Analysis: Pandas and NumPy, Coursera Project Network
- Getting Started with Power BI Desktop, Coursera Project Network